SDG BLOCKCHAIN ACCELERATOR

Impact Measurement Framework

1. Project Information

- **Project Name:** Digital Infrastructure for Green Energy Exports / Fuel Switch x UNDP BiH Pilot – Cohort 2
- **Challenge & UNDP Office:** UNDP Bosnia and Herzegovina – Energy Sector Decarbonization & CBAM Compliance

- **Document Version:** 1.0 (Final Submission)

2. SDG Alignment

(Specify which UN Sustainable Development Goals your project addresses. Include the specific targets and indicators.)

SDG 7: Affordable and Clean Energy

- Target 7.2: By 2030, substantially increase the share of renewable energy in the global energy mix.

- Indicator: Volume of renewable energy (MWh) certified and monetized via the platform, incentivizing new production capacity.

SDG 13: Climate Action

- Target 13.2: Integrate climate change measures into national policies, strategies, and planning.

- Indicator: Number of verified Guarantees of Origin (GOOs) retired for CBAM compliance, serving as proof of low-carbon exports.

SDG 9: Industry, Innovation, and Infrastructure

- Target 9.4: Upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency.

- Indicator: Reduction in carbon intensity (CO_2e/unit) of BiH export products (electricity/commodities) verified through blockchain traceability.

3. Key Performance Indicators (KPIs)

(Define measurable indicators that show your project's impact.)

KPI	Measurement Method	Target / Threshold	Notes
Volume of Energy Tokenized	Sum of MWh values in minted token metadata	**100 - 500 MWh** (Pilot Phase)	Represents the real-world green energy assets brought on-chain from PV Romanovci.
Successful On-Chain Verifications	Cardano Testnet Transaction Logs	**100% success rate** for mint/transfer/burn	Every certificate must have a verifiable transaction hash to be valid for CBAM.
Pilot Partner Adoption	Active Accounts on Platform	**2 Tier-1 Partners** (UNDP, ELIFA-SOIL)	Confirmed onboarding of the Producer and Offtaker/Trader.
Compliance Reports Generated	Backend System Logs & Off-chain PDF Audit	**1 Validated CBAM Report**	The final output used by the exporter to prove origin to regulators.

4. Tracking Methods

(Describe how you will monitor and collect data for KPIs.)

On-chain Metrics:

- Minting Events: Monitoring the specific Policy ID on the Cardano blockchain to track the creation of new GOO tokens.
- Metadata Validation: Using a block explorer (e.g., Cardanoscan) to verify that the JSON metadata (Producer ID, MWh, Date) is correctly anchored to each transaction.
- Token Retirement (Burning): Tracking transactions sent to the burn address to confirm certificates have been "used" and cannot be double-counted.

Off-chain Metrics:

- Smart Meter Data Ingestion: Automated logs verifying the receipt of generation data from the producer (PV Romanovci) or Data Administrator (OIEiEK).
- User Dashboard Analytics: Asset registrations, and asset retirement requests via the Fuel Switch admin panel.
- Stakeholder Feedback: Qualitative feedback from UNDP and ELIFA-SOIL on the usability of the generated compliance reports.

5. Baseline vs Projected Outcomes

(State your starting point and what you expect to achieve.)

Metric	Baseline	Projected Outcome
Verifiable Digital GOOs	0 (Manual/Paper-based)	**Fully functional digital registry** on Cardano
Traceability of Exports	Fragmented / Siloed Data	**100% Immutable Audit Trail** for pilot volume
CBAM Compliance Readiness	Low / High Risk of Tax	**High / Technical Readiness** demonstrated
Data Transparency	Private / Opaque	**Publicly Verifiable** via Blockchain Explorer

6. Monitoring & Reporting Plan

(Define who is responsible for reporting, how often, and what format.)

Frequency: Weekly technical checks during the active pilot; Monthly impact reporting post-pilot.

Responsible Party:

- Technical Monitoring: Fuel Switch Developer (Johandré Hoon) – Responsible for smart contract health and on-chain logs.
- Impact Reporting: Fuel Switch Project Lead (Zandré Zandberg) – Responsible for consolidating partner feedback and export metrics.

Format:

- Technical: Automated system logs and transaction hash lists.
- Strategic: "Pilot Impact Report" (PDF) summarizing volumes, costs savings, and user feedback.

Tools: Cardanoscan (Block Explorer), Fuel Switch Admin Dashboard, Google Data Studio.

7. Risks & Mitigation

(Optional: outline potential risks in measuring impact and how you will address them.)

Risk: Data Oracle Disconnect.

- Description: Failure to receive automated meter data from the grid operator (OIEiEK) due to API issues.
- Mitigation: Implementation of a manual "verified upload" feature where signed CSV generation reports can be uploaded and hashed to the blockchain as a fallback data source.

Risk: Regulatory Interpretation.

- Description: EU or local regulators (DERK) questioning the legal validity of a blockchain token as a certificate.
- Mitigation: The platform generates a standard PDF retirement report backed by the blockchain data. We present the PDF (a familiar format) to regulators, using the blockchain hash only as the background "digital notary" seal.

Risk: Privacy Concerns.

- Description: exposing sensitive commercial trade data on a public ledger.
- Mitigation: We use a "hashed reference" model. Sensitive commercial data (prices, specific contract terms) remains off-chain in the private database; only the verification proofs (energy volume, origin, unique ID) are published on-chain.

Risk: Institutional Inertia & Political Gridlock.

- Description: The complex governance structure in BiH (State vs. Entity levels) may cause significant delays in securing formal endorsements or regulatory approvals from bodies like MOFTER or DERK, potentially stalling the transition from pilot to national adoption.
- Mitigation: "Private-First" Execution Strategy. We focus pilot execution on agile private sector partners (ELIFA-SOil, PV Romanovci) to generate verifiable data and economic proof-points before seeking legislative changes. Simultaneously, the UNDP leverages its neutral diplomatic status to facilitate high-level policy dialogue, insulating the technical pilot from political deadlocks.